

Theory 5.1.1

projections: $\vec{p} \cdot \vec{v} = (x, y) \cdot (1, -1) = \underline{x - y}$

triangle: $\left. \begin{array}{l} (2, 2) \rightarrow 2 - 2 = 0 \\ (5, 3) \rightarrow 5 - 3 = 2 \\ (3, 6) \rightarrow 3 - 6 = -3 \end{array} \right\} \{-3, 0, 2\}$

quad: $\left. \begin{array}{l} (7, 2) \rightarrow 7 - 2 = 5 \\ (10, 3) \rightarrow 10 - 3 = 7 \\ (6, 5) \rightarrow 6 - 5 = 1 \\ (9, 6) \rightarrow 9 - 6 = 3 \end{array} \right\} \{1, 3, 5, 7\}$

$\Rightarrow$ 1D-Intervals: $\textcircled{A} [-3, 2]$ $\textcircled{B} [1, 7]$

$\approx$ overlap in interval $[1, 2]$, collision can not be ruled out.

5.1.2

Ray $x(t) = p + t \cdot d$

$p = \begin{pmatrix} 1 \\ 0,2 \\ 0,5 \end{pmatrix}$

$h = 0,3$

$\hat{d} = \frac{1}{\sqrt{0,3^2 + 0,5^2 + 1^2}} \cdot \begin{pmatrix} 0,3 \\ 0,5 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{1,34}} \begin{pmatrix} 0,3 \\ 0,5 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 0,26 \\ 0,43 \\ 0,86 \end{pmatrix}$

start cell (i_x, i_y, i_z) : $i_\alpha = \left\lfloor \frac{p_\alpha - g_{\min, \alpha}}{h} \right\rfloor$

$i_x = \left\lfloor \frac{1}{0,3} \right\rfloor = 3$

$\rightarrow g_{\min, \alpha} = (0, 0, 0)$

$i_y = \left\lfloor \frac{0,2}{0,3} \right\rfloor = 0$

$i_z = \left\lfloor \frac{0,5}{0,3} \right\rfloor = 1 \quad \Rightarrow \underline{\text{start cell } (3, 0, 1)}$

1. Intersection

since $\hat{d}_x > 0$:

$x_{\text{next}} = (i_x + 1) h = (3 + 1) \cdot 0,3 = 1,2$

$y_{\text{next}} = (i_y + 1) h = 0,3$

$$y_{\text{next}} = (i_y + 1)h = 0,3$$

$$z_{\text{next}} = (i_z + 1)h = 0,6$$

$$t_{\text{max},x} = \frac{x_{\text{next}} - p_x}{d_x} ; \begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix} \approx \begin{pmatrix} 0,26 \\ 0,43 \\ 0,86 \end{pmatrix} ; p = \begin{pmatrix} 1 \\ 0,2 \\ 0,5 \end{pmatrix}$$

$$t_{\text{max},x} = \frac{1,2 - 1}{0,26} \approx 0,77$$

$$t_{\text{max},y} = \frac{0,3 - 0,2}{0,43} \approx 0,23$$

$$t_{\text{max},z} = \frac{0,6 - 0,5}{0,86} \approx 0,12$$

$$\left. \begin{array}{l} t_1 = \min(0,77; 0,23; 0,12) \\ t_1 = t_{\text{max},z} = 0,12 \end{array} \right\}$$

~ ray exits through z-plane
 $\Rightarrow$ increment i_z : cell 2 (3,0,2)

$$\text{Step increments } \Delta t_x = \frac{h}{d_x}$$

$\Rightarrow$ Since stepping in z direction, update only $t_{\text{max},z}$

$$t_{\text{max},z} \leftarrow t_{\text{max},z} + \Delta t_z = 0,12 + \frac{0,3}{0,86} \approx 0,46$$

updated $t_{\text{max},x}$ -values:

$$t_{\text{max},x} \approx 0,77$$

$$t_{\text{max},y} \approx 0,23$$

$$t_{\text{max},z} \approx 0,46$$

$$\left. \begin{array}{l} \text{be minimum ...} \\ t_2 = t_{\text{max},y} \approx 0,23 \end{array} \right\}$$

~ y-plane exit, incr. i_y : cell 3 (3,1,2)

$\rightarrow$ update $t_{\text{max},y}$

$$t_{\text{max},y} \leftarrow t_{\text{max},y} + \Delta t_y = 0,23 + \frac{0,3}{0,43} \approx 0,93$$

~ $t_{\text{max},z} = 0,46$ is new minimum.

~ z-plane exit ~ cell 4 (3,1,3)

First four cells: $(3,0,1) \rightarrow (3,0,2) \rightarrow (3,1,2) \rightarrow (3,1,3)$

first steps:

$$t_1 \approx 0,12 \quad , \quad t_2 \approx 0,25$$